Benchmark: **MA.7.GR.1.3** · Circumference of a circle Student code: _____ Calculator: yes

Learning goal: I can use the relationship between circumference and diameter. (circle your level after scoring)

4	I can explain pi as the ratio of circumference to diameter and use it in a real-world problem.
3	I can find circumference from radius or diameter, and radius or diameter from circumference.
2	I can identify the radius, diameter, and circumference of a circle.
1	I need help to start this benchmark.

1. A circle has a radius of 11 centimeters.

What is the circumference of the circle? Write your answer in terms of π.

2. The circumference of a round pool is 106.81 feet.

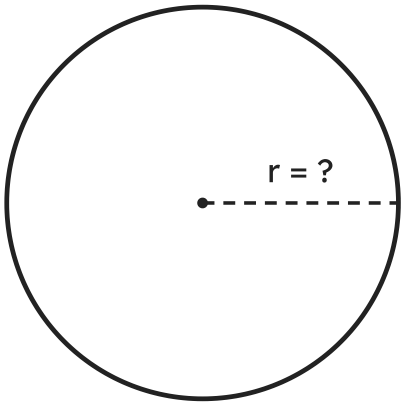
What is the diameter of the round pool? Round your answer to the nearest hundredth. Use the π key on your calculator.

Turn over →

Benchmark: **MA.7.GR.1.3** · Circumference of a circle Student code: _____

3. A circular clock face has a circumference of 62.80 centimeters, as shown.

What is the radius of the circular clock face? Round your answer to the nearest hundredth. Use 3.14 for π.



C = 62.80 cm

Not drawn to scale

4. A circular tabletop has a radius of 2 feet.

What is the circumference of the tabletop? Round your answer to the nearest hundredth. Use the π key on your calculator.

5. A scooter wheel has a diameter of 12 inches. The wheel rolls in a straight line for 200 feet.

How many complete turns does the wheel make? ($1 \text{ foot} = 12$ inches) Use 3.14 for π.

Score: ____ / 5 **My level: 1 2 3 4** **→ copy both to your progress strip**

Answer key — C-GR.1.3 · MA.7.GR.1.3 · Circumference of a circle

Unit 4: Area. Regular = on-benchmark item; Rigor = the item that needs the benchmark's reasoning. Mastery on the weekly check = 4 of 5.

Weekly check (5 items)

1. Regular **22π centimeters**

$C = 2\pi r = 2 \times \pi \times 11 = 22\pi$ centimeters.

2. Regular **34 feet (34.00 also accepted)**

$C = \pi d$, so $d = C \div \pi$.

With the π key: $106.81 \div \pi \approx 34.00$ feet.

3. Regular **10 centimeters (10.00 also accepted)**

$C = 2\pi r$, so $r = C \div (2\pi)$.

$2 \times 3.14 = 6.28$, and $62.80 \div 6.28 = 10.00$ centimeters.

4. Regular **12.57 feet**

$C = 2\pi r = 2\pi(2) = 4\pi$.

With the π key: $4 \times \pi \approx 12.57$ feet.

5. Rigor **63 complete turns**

In one turn, the wheel rolls a distance equal to its circumference: $C = \pi d = 3.14 \times 12 = 37.68$ inches.

$200 \text{ feet} = 200 \times 12 = 2400 \text{ inches}$.

$2400 \div 37.68 \approx 63.69$ turns.

The wheel makes 63 complete turns; the last turn is not finished.

Re-check strip (3 items, C-GR.1.3-R)

1. Regular **13 centimeters (13.00 also accepted)**

$C = 2\pi r$, so $r = C \div (2\pi)$.

With the π key: $81.68 \div (2\pi) \approx 13.00$ centimeters.

2. Regular **3.14, which is an approximation of π**

The constant of proportionality is circumference ÷ diameter.

$25.13 \div 8 \approx 3.14$

$43.98 \div 14 \approx 3.14$

$62.83 \div 20 \approx 3.14$

The ratio is the same for every circle. That constant is π, and 3.14 is the common approximation of it.

3. Regular **30 feet**

$C = 2\pi r$, so $60\pi = 2\pi r$.

Divide both sides by 2π: $r = 60 \div 2 = 30$ feet.

Comparison strip (2 items, C-GR.1.3-X) — used on a later week's re-check when this benchmark was *not* retaught

1. Regular **36π inches**

$C = 2\pi r = 2 \times \pi \times 18 = 36\pi$ inches.

2. Regular **8 centimeters (8.00 also accepted)**

$C = 2\pi r$, so $r = C \div (2\pi)$.

$2 \times 3.14 = 6.28$, and $50.24 \div 6.28 = 8.00$ centimeters.

Reteach plan — C-GR.1.3

MA.7.GR.1.3 · Circumference of a circle

Item that gave the most trouble (from both tally sheets):

What the mistake looks like:

Worked example for Tuesday (8–10 min):

Three practice items (one change at a time):

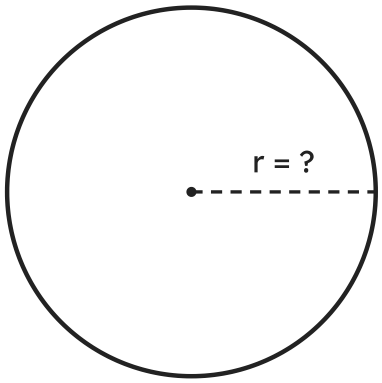
Decision line: Reteach on _____ Huddle: in person / paper Date: _____ Re-check given on _____

7th Grade Math · Re-check (3 items) C-GR.1.3-R

Benchmark: **MA.7.GR.1.3** · Circumference of a circle Student code: _____ Calculator: yes

1. A round plate has a circumference of 81.68 centimeters, as shown.

What is the radius of the round plate? Round your answer to the nearest hundredth. Use the π key on your calculator.



C = 81.68 cm

Not drawn to scale

2. A student measured the diameter and the circumference of three paper cups, in centimeters. The results are shown in the table.

Diameter (cm)	Circumference (cm)
8	25.13
14	43.98
20	62.83

The circumference of a circle is proportional to its diameter. Based on the table, what is the constant of proportionality, rounded to the nearest hundredth?

3. The circumference of a circle is 60π feet.

What is the radius of the circle?

Score: ____ / 3 → copy to your progress strip

7th Grade Math · Comparison items (2) C-GR.1.3-X

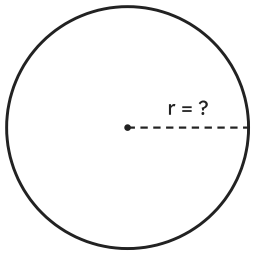
Benchmark: **MA.7.GR.1.3** · Circumference of a circle Student code: _____ Calculator: yes

1. A circle has a radius of 18 inches.

What is the circumference of the circle? Write your answer in terms of π.

2. A circular clock face has a circumference of 50.24 centimeters, as shown.

What is the radius of the circular clock face? Round your answer to the nearest hundredth. Use 3.14 for π.



C = 50.24 cm

Not drawn to scale

Score: ____ **/ 2** → **copy to your progress strip**

7th Grade Math · Comparison items (2) C-GR.1.3-X

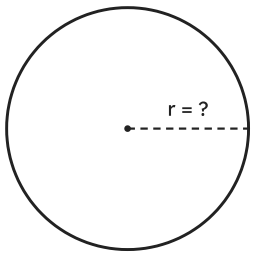
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What is the radius of the circular clock face? Round your answer to the nearest hundredth. Use 3.14 for π.



C = 50.24 cm

Not drawn to scale

Score: ____ **/ 2** → **copy to your progress strip**